# Minglang Chen

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## Skills

**Languages**: Java, C, C++, Python, JavaScript, TypeScript, HTML, CSS, Bash, SQL
**Frameworks & Libraries**: React, React Native, Expo, Tailwind CSS, Node.js, Express, NumPy, Pytest, Qiskit
**Tools & Platforms**: Linux, Git, Supabase, Vite, Vercel, Autodesk Fusion, Claude Code

## Education

**University of Waterloo** **Sep 2026 – Present**
*Bachelor of Software Engineering (Honours, Co-op) | President's Scholarship of Distinction ($2,000)* *Waterloo, ON*

## Projects

**Leavy: Calendar-Based Departure Alerts** **Personal Project**
*TypeScript · React Native · Expo · Supabase · Google Maps API · GitHub Actions* *Mobile, Web | leavy-web.vercel.app*

- Built a mobile and web app that pairs calendars with live Google Maps travel times and a grace period to tell users when to leave, helping them stop arriving late
- Added Google and Apple sign-in with Supabase so preferences sync across phone and web, plus a no-account guest mode
- Automated checks on every pull request by setting up GitHub Actions for lint, type checks, and Claude PR reviews

**Autonomous Line-Following Vehicle** **University Hill Secondary School**
*Arduino · C++ · Embedded Hardware* *github.com/langchengit/engineering-car-project*

- Programmed an Arduino self-driving car to follow tape paths and avoid obstacles with IR and ultrasonic sensors, building my foundation in embedded hardware
- Controlled wheel speed and direction through a motor driver, and designed and soldered the full circuit

**Zoetrope** **University Hill Secondary School**
*Arduino · C++ · Fusion 360 · 3D Printing* *Vancouver, BC*

- Built a team zoetrope that flashes LEDs in sync with a spinning flywheel, making 3D printed frames look animated
- Slowed the flywheel and raised its torque by designing compound gears and 3D printing gears and stands
- Modelled a hall sensor holder in Fusion 360 to measure flywheel RPM, keeping the LED flashes in sync with its speed

## Experience

**vsHacks** vshacks.com **Sept 2024 – Present**
*Vice Chair | Outreach, Web Development, GitHub, Event Operations* *Vancouver, BC*

- Co-organized a two-day online high school hackathon that grew from **100+** (2025) to **400+** (2026) participants, running outreach and the Devpost portal
- Led sponsorship outreach to companies, raising **$7K+** across both years
- Automated sign-up verification and role assignment for **400+** participants by building a Discord bot
- Taught beginners Git through a GitHub workshop, and hosted a career session with a Microsoft ML Engineer

**University Hill Secondary School Physineering Club** **Sept 2025 – June 2026**
*President | Physics, Engineering, Teaching, Competition Prep* *Vancouver, BC*

- Ran weekly physics and engineering lectures built on hands-on research, like taking apart a Pixel phone and projector
- Led teams at **3** competitions (FYKOS, Kwantlen Science Challenge, UBC Physics Olympics) through regular practice

**Learning Buddies Network** **Oct 2025 – May 2026**
*Math Mentor (Volunteer) | Mathematics, Tutoring, Communication, Lesson Planning* *Remote*

- Tutored an elementary student **twice a week**, improving subtraction, division, and word problems with planned lessons
- Introduced decimals, fractions, and variables early, adapting my explanations after lesson reviews with mentors

## Awards & Competitions

**Programming**: USACO Silver Division; Canadian Computing Competition: Junior Honor Roll (full marks, 2023), Senior Top 25% (2024)
**Mathematics**: Euclid Contest Top 25% (2024, 2025); AIME participant; CSMC Honor Roll (2023, 2025); Cayley Contest Honor Roll (2023); Fermat Contest Honor Roll (2024); Stanford Math Tournament (team Top 40%)
**Physics**: UBC Physics Olympics: 2nd place, Quizzics (2024, 2025)